

4. Fertilizers and Their Use in Bangladesh

Broadly a fertilizer may be defined as any substance (chemical, organic and microbial) that is added to soil to supply element(s) required for the nutrition of plants. Specifically, fertilizers are chemical, organic or microbial materials that occur naturally or are produced in a factory and when added to the soil, they supply nutrient elements required for normal plant growth.

4.1 Types of fertilizer

Inorganic (Mineral) fertilizer: Fertilizer which contains nutrients in the form of inorganic salts obtained by extraction and/or by physical and/or chemical industrial processes, e.g. TSP, MOP, gypsum etc.

Organic fertilizer: Carbonaceous materials mainly of plant and/or animal origin added to the soil specifically for the nutrition of crops, e.g. cowdung, poultry manure, compost etc.

Straight fertilizer: Fertilizer which contains only one primary nutrient (N, P or K), e.g. urea, TSP and MOP.

Micronutrient fertilizer: Fertilizer which contains any micronutrient (Zn, B, Fe, Mn, Cu, Mo, Cl or Ni), which is required by plants relatively to a smaller amount but essential for plant growth, e.g. zinc sulphate, boric acid, copper sulphate etc.

Compound fertilizer: Fertilizer which contains two or more nutrient elements, e.g. DAP (18%N, 20%P).

Granular fertilize: Solid material that is formed into a definite sized granule, e.g. USG, UMG.

Coated fertilizer: Granular fertilizer that is coated with a thin layer of different materials in order to improve the behavior and/or modify the characteristics of the fertilizer, e.g. neem coated urea. Coated fertilizers are slow release fertilizers.

Liquid fertilizer: Fertilizer that is in suspension or solution, e.g. liquor ammonia.

Nanofertilizers- Fertilizers that are of nano sizes particles (below 100 nm).

Biofertilizer: Fertilizer that contains an active culture of beneficial microorganisms which benefit the plants by providing N or P or rapid mineralization of organic material, e.g. *Rhizobium*, *Azotobacter*, *Azospirillum* etc.

Plant Growth Regulator (PGR): Plants along with sunlight, water, oxygen and mineral nutrients also require certain organic compounds to signal, regulate and control this growth. These organic substances are collect PGR e.g. auxin, cytokinins, gibberellins, ethylene etc.

Apart from fertilizers, some materials are used in soil as amendments:

Soil conditioner: Material that added to soils for improvement of their physical and/or chemical properties and/or biological activity.

Liming material: An inorganic soil conditioner containing Ca or Ca and Mg, generally in the form of oxide, hydroxide, or carbonate, principally intended to raise the pH of a soil. Dolomite [$\text{CaMg}(\text{CO}_3)_2$] is commonly used in Bangladesh.

4.2 Nanofertilizer

Nanofertilizers are developed by the use of nanotechnology. Use of nanofertilizers has many benefits such as improving nutrient use efficiency (N, P, Zn etc.), reducing nutrient loss (erosion, leaching), maintaining soil health, lowering environmental pollution (heavy metals) and thus increasing crop yield. Nano size particles of nanofertilizers (size below 100 nm) provide a high surface area-to-volume ratio that enables plants to take up nutrients slowly and sustainably as needed. One nanometer is one-billionth of a meter i.e. 10^{-9} .

Nano fertilizers may be considered as new frontiers of modern agro-inputs and can revolutionize the field of soil fertility and fertilizer management in Bangladesh. Many researchers have formulated nanofertilizers by incorporating hydroxyapatite $\{Ca_{10}(PO_4)_6(OH)_2\}$ and urea $\{CO(NH_2)_2\}$ to increase delivery of nutrients to the plants. It is evidenced that hydroxyapatite (HA) has many biomedical and agricultural applications because of its excellent bioactivity and biocompatibility.



Nanotechnology has brought numerous developments in the agricultural sector through green nanomaterials, nanofertilizers, nanoagrochemicals, nanopesticides, nanoherbicides, nanobiosensors, and nano-based treatment of agricultural wastes to increase the food productivity by foliar and soil application as well as to establish the sustainable agricultural management. Nano-trapped fertilizer filled with biological agents (microorganisms, enzymes) can render long-term profits for crop production.

Although the use of nanofertilizers is a new approach towards smart and sustainable agriculture, their potential threat to plants, soil organisms, and human health should also be carefully considered before commercial application. Accumulation of nanomaterials in the environment and food chain may pose a risk to human health.

4.3 Deep placement of urea

Urea Super Granules (USG) and Urea Mega Granules (UMG) are used to increase nitrogen use efficiency. Urea Super Granules (USG) are small sized granules, each weighing 0.9g or 1.8g and Urea Mega Granules (UMG) are large sized granules, each 2.7g pellets made of ordinary prilled/granular urea by compressing. The amount of USG or UMG should be adjusted to the recommended dose of N for different crops and soils. The granules (USG/UMG) need to be placed after week of transplanting of rice at 8-10 cm soil depth at the centre of four hills between rows 1 and 2, between rows 3 and 4, and so on. It is possible to save 30% urea-N by using USG or UMG.

With USG, recovery of deep placed N in wetland rice is greater than the N recovery from surface applied and/or incorporated ordinary urea. Fertilizer recovery in the wetland rice plant tops was found significantly higher for deep placed N as USG/UMG/ ^{15}N (50-60%) than for split-applied Urea ^{15}N (25-34%). About 35% N can be saved by using USG or UMG in rice and 20% in vegetable and fruit crops (viz. cabbage, cauliflower, tomato, potato and papaya).

The main benefit of USG/UMG placement is that the N loss through NH_3 volatilization, denitrification, leaching and surface runoff are significantly minimized. Deep-placed N as USG/UMG is less subject to algal immobilization and uptake by aquatic weeds than broadcast and/or incorporated urea. These two factors contribute to the improved nitrogen use efficiency.

4.4 Biofertilizers

Biofertilizers are microbial inoculants consisting of living and active strains of specific bacteria, algae, fungi, alone or in combination, used for application to seed, soil or composting areas with the objective of increasing crop productivity. They help in the biological nitrogen fixation, solubilization of insoluble phosphate, stimulating plant growth or in decomposition of organic substances.

Rhizobial biofertilizer

Rhizobial biofertilizer/inoculants are made with bacterial strains that can fix atmospheric N₂ in symbiosis with legumes. They are the members of five bacterial genera: *Rhizobium*, *Mesorhizobium*, *Bradyrhizobium*, *Sinorhizobium* and *Azorhizobium*. The beneficial effect of these organisms in increasing yield of leguminous crops (lentil, chickpea, cowpea, mungbean, blackgram, pigeonpea, grasspea, pea, groundnut and soybean) from fixation of atmospheric nitrogen is well established.

Methods of application of rhizobial biofertilizers

- i) Take an amount of seed in a container/bowl.
- ii) Add sugarcane molasses (2-3% for large seeds viz. groundnut, soybean etc. and 3-5% for small seeds viz. lentil, mungbean, etc.) and mix with seeds to make them sticky.
- iii) Add peat based inoculant (2-3% for large seeds and 3-5% for small seeds) to sticky seeds.
- iv) Mix seeds with inoculum until they are coated and appear uniformly black.
- v) Dry seeds under shade on a paper. Do not dry in direct sunlight.
- vi) There should be a minimum 24-hour gap between seed treatment with fungicide and biofertilizer application.
- vii) Use double quantity of biofertilizer in case of pesticide treated seeds.
- viii) Sow the inoculated seeds and cover the seeds with soil immediately.

Azolla biofertilizer

Azolla is a floating fern and it fixes atmospheric N₂ in symbiosis with *Anabaena azollae*. In wet land rice, the average N contribution by Azolla is equivalent to 25-40 kg N/ha. Azolla should be applied to standing water @ 3-4 t/ha at 1-2 weeks after transplanting of rice. After 2-3 weeks of its application, water needs to be drained out and Azolla should be incorporated into the soil. Azolla can be grown in the rice field more than once.

4.5 Time and methods of fertilizer application

Fertilizer use efficiency depends to a great extent on the time and methods of fertilizer application. This is particularly important for nitrogenous fertilizers like urea, which is highly water soluble and easily lost from soil in various ways. As much as 70% of the applied urea might be lost from soil under wet land rice culture.

In fertilizer application, the following principles should be taken into account.

- Fertilizers should not be applied too close to seeds, young roots and stems of plants
- Fertilizers should not be applied to plant leaves when they are young and wet.

- Manure and fertilizers should be mixed thoroughly with the soil.
- Urea should not be applied to soil with standing water.
- Manure should be applied 7-10 days before sowing/planting.

For efficient use of fertilizers, an appropriate method of fertilizer application is very important. There are three methods of fertilizer application: broadcast, localized application and foliar spray. Broadcast method is most commonly used in field crops and the localized method is widely followed in horticultural crops. Foliar application could be effective for correcting deficiency of micronutrients (Fe, Mn, Zn and B), not for that of any macronutrient. Maximum absorption of foliar applied nutrient occurs when (i) the nutrient element is in low concentration, (ii) applied solution contains a wetting agent, (iii) plant tissue is fully turgid and at the mature stage, (iv) applied in the late afternoon or in the morning when dew dries up, and (v) air temperature is cool, has a high relative humidity, and there is no wind.

In order to increase the efficiency of fertilizers, the following guidelines need to be followed.

- During rabi season, if there is no facility for irrigation, the full dose of urea should be applied and mixed with the soil during final land preparation. If irrigation facility is available, urea should be applied in three equal splits, during final land preparation, rapid vegetative growth stage, and 5-7 days before primordial initiation.
- In case of rice (except B. Aman rice), urea should be applied in three equal splits, the first split at final land preparation or immediately after seedling establishment, the second split at rapid tillering stage and the third split at 5-7 days before panicle initiation (PI) stage. In case of broadcast rice culture, nitrogen fertilizer should be applied in two equal splits, the first one as basal and the second one at maximum tillering stage.
- For vegetables, 2-3 split applications of urea should be adjusted with the growth periods of the crop. For short duration crops the full dose of urea may be applied during final land preparation.
- For most spices crops, urea application may be made in 2-3 splits.
- Phosphate fertilizer should be applied 1-2 days prior to final land preparation and zinc fertilizer should be applied during final land preparation.
- The full dose for potassium, sulphur, magnesium and boron fertilizers should be applied as basal during final land preparation. In coarse-textured soils, the potassium fertilizer may be applied in two splits, the one-half at final land preparation and the rest half at rapid tiller stage. Zinc and P fertilizers should not be mixed during storage.
- Topdressing should be done under moist soil condition and mixed thoroughly with the soil as soon as possible for better utilization.
- For onion, potassium should be applied as K_2SO_4 instead of KCl (MOP).
- Micronutrient fertilizers can be applied as foliar spray since these are required by crops relatively in a smaller amount. Foliar spray has some advantages: (i) Application rate is much lower than soil application rate, (ii) Uniform application is possible, (iii) Response to the applied nutrient is almost immediate. Nevertheless, it has some disadvantages: (i) Leaf burn may result if salt concentration is too high, (ii) Nutrient demand may not be fulfilled when the plants are small and leaf surface is insufficient for foliar absorption, (iii) There is no residual effect.

The sources and solution concentration of different micronutrients could be as follows:

Micronutrient	Source	Concentration (g/100 litre)
Iron	Ferrous sulphate	250-350
Manganese	Manganese sulphate	250-500
Zinc	Zinc sulphate	250-600
Copper	Copper sulphate	250-600
Boron	Solubor	250-600
Molybdenum	Ammonium molybdate	30-60

4R Nutrient Stewardship of IPNI (International Plant Nutrition Institute) is the foundation of best management practices for fertilizer management. The 4R represents right source (types of fertilizer), right rate (rate of application), right time (basal & stage of crop growth) and right place (method of application)

4.6 Rationale of fertilizer use

Except nitrogenous fertilizers, all fertilizers would have residual effect on the next crops. Thus, application of those fertilizers at a full rate is not economical. Considering nutrient uptake, fertilizer types, crop types and soil characteristics the following points need to be considered while making fertilizer recommendations for a cropping pattern.

- i) More than 60% of added N is lost from soil in different ways (denitrification, volatilization and leaching) and very little or no residue is left into the soil for use by the next crops. Thus, urea fertilizer should be applied at a full rate to every crop. The N rate for T. Aman rice could be reduced by 25-30 kg ha⁻¹ if 12-15 tons green biomass of dhaincha (*Sesbania*) is incorporated. This reduction could be 8-10 kg ha⁻¹ when stover of grain legumes (e.g. mungbean) is incorporated to soil.
- ii) Phosphorus availability is low in both acid and calcareous soils. Plant uptake of this element is usually 15-25% of the added amount, thus it produces residual effect on the subsequent crops. Hence, the P rate for the second and third crops could be reduced by 40-50% for rice and jute, and by 30-40% for vegetables.
- iii) Potassium availability is low in light textured, terrace and piedmont soils. The K requirement for rice, tuber, jute, sugarcane, fruit and vegetable crops is high. Potassium application should be considered as a maintenance dose even at optimum level of soil K. Potassium application can be reduced by 30-40% in the subsequent crops after potato, sugarcane and vegetables when recommended dose of K fertilizer was used. The K dose could be reduced by 20-40% in subsequent crops if 2-4 tons of crop residues per hectare are incorporated to soils.
- iv) Sulphur availability is low under wetland condition. The S fertilizer (e.g. gypsum) has substantial residual effect on the next crops. So, wetland rice should receive full dose of S. Cultivation of rabi crops after harvest of T. Aman rice should receive 50% of the recommended dose of S except for oil seed, maize, potato and vegetable crops, which should receive full dose of S. For 3-rice and 3-vegetables cropping patterns, the 2nd and 3rd crops would receive 50% S dose.

- v) Magnesium availability is generally low in piedmont and Tista floodplain soils. Magnesium fertilizer should be applied to rabi crops and also to kharif crops e.g. jute. For wet land rice cultivation, magnesium application is not needed. If dololime ($\text{CaCO}_3 \cdot \text{MgCO}_3$) is used to raise soil pH, application of Mg fertilizer is not needed within three years.
- vi) Zinc availability is low in calcareous and wetland rice soils. Zinc fertilizers should be applied to both rabi and kharif crops when grown in calcareous soils (AEZs 10, 11, 12 and 13). For the rice-rice cropping pattern, full dose of Zn should be applied to the 1st crop and 50% dose to the 2nd crop. There is no need to apply Zn fertilizer to the rabi crops when the preceding T. Aman rice has received Zn at a full rate. Maize, potato, vegetables and spices as the 2nd or 3rd crop should receive 50% rate of Zn.
- vii) Boron availability is low under dry land condition. Rabi crops should receive boron once a year. In the cropping pattern where rabi is the fallow period then full recommended dose of boron should be applied to the kharif-1 crop.
- viii) If organic manure such as cowdung, poultry manure, bioslurry and compost are used, the amount of nutrients from manure should be adjusted with the rates of chemical fertilizers, taking into account of 50% mineralization.
- ix) Under rainfed condition, the yield reduction could be 15% for rice and jute, 20% for potato and sugarcane, and 35% reduction for wheat, oilseed, vegetables and spices, for which all the recommended fertilizer nutrients (N, P, K, S, Mg, Zn and B) should be reduced by 25% compared to that with irrigated condition.
- x) For an intercropping system, fertilizer dose for the companion crop should be 20-50% of the recommended dose for the crop depending on population and the main crop should receive 100% recommended dose of fertilizers.

4.7 Crop response to added fertilizers

Based on soil test result, nutrient content can be classified into five categories- very low, low, medium, high and very high. The probability of crop response to applied nutrients increases with decreasing soil test value. There is a good possibility of obtaining profitable crop response from the use of fertilizers on soils showing “low to medium” status of a nutrient. There is a greater possibility of obtaining a response from a given nutrient with a low soil test result.

Soil test values would be of no value unless they are positively correlated with crop response. Yield responses from applied nutrients can be more closely related to the status of nutrient availability in the soil as determined by soil analysis. A soil test measures part of the total nutrient status in the soil and represents only an index of nutrient availability. Soil tests do not measure the exact quantity of a nutrient potentially taken up by a crop. To predict the nutrient needs of a crop, the soil test values must be calibrated against nutrient rate that used in the field experiments.

Crop growth response curve

The response of a crop to added nutrients shows a well-defined pattern. When any growth parameter or yield of a crop is plotted as a function of increasing amounts of an applied nutrient, successive increments of nutrient give successively smaller increases in crop growth or yield. Such curves are known as growth response curves (Fig. 3). From the response function equation, the value of an added nutrient that maximizes crop yield is estimated as follows:

$$Y = a + bx + cx^2$$

Where, Y is the maximum crop growth or yield

a, b and c are co-efficients and

x is the added nutrient

The optimum nutrient rate (Ny) can be computed from the above equation as

$$N_y = \frac{-b}{2c}$$

Now, the nutrient rate that maximizes profit:

$$N_p = \frac{1}{2c} \left[\frac{P_f}{P_y} - b \right]$$

Where, P_f = Price of nutrient, P_y = Price of crop product

Response of a crop to an added nutrient depends on soil-crop-climatic variables and production practices. In general, most of the crops are responsive to N fertilizer. Upland and rabi crops grown under strongly acidic and alkaline soil conditions are more responsive to P fertilizers. Oil seed crops and wetland rice are responsive to S fertilizers. Sugarcane and tuber crops are more responsive to K fertilizers. Crops grown on light textured soils respond more to K fertilizers. Crops grown on the calcareous and saline soils respond more to Zn fertilizers. Fertilizer response is more pronounced under irrigated than under rainfed ecosystems. Crops may differ markedly in their requirements of different nutrients. Nonetheless, soil moisture content may affect the response of a crop to fertilizers. Furthermore, fertilizer response of a crop depends on the kinds of fertilizers used, and also on the time and methods of their application. To obtain the maximum benefits from fertilizer use 4R Nutrient Stewardship should be followed – Right source, Right rate, Right time and Right place

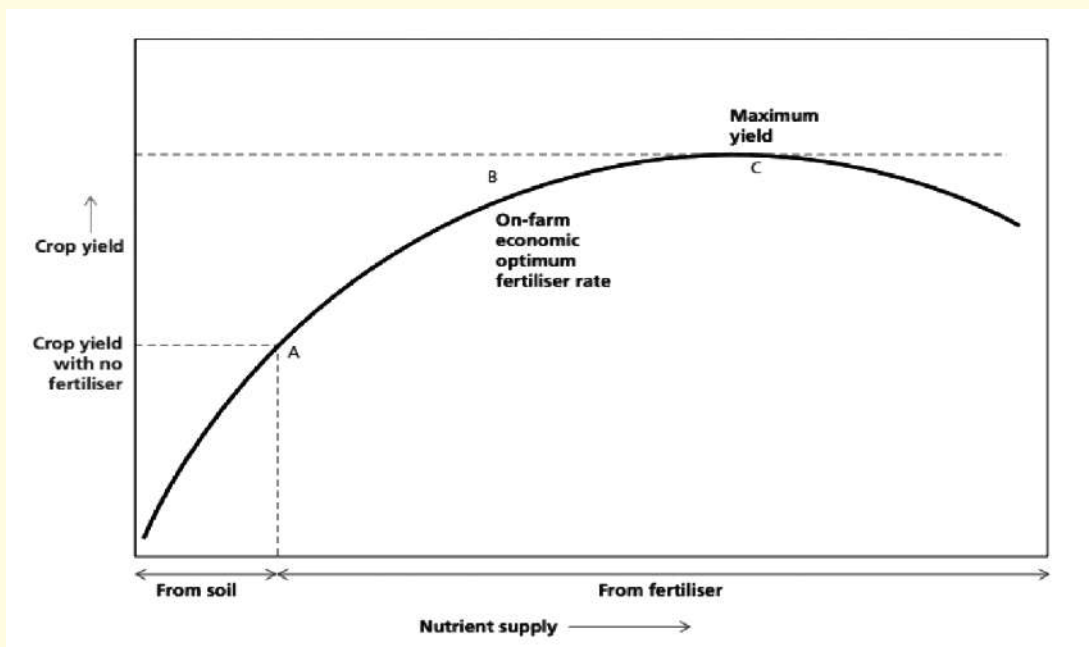


Fig. 3 Crop yield as influenced by nutrient supply from soil and fertilizer sources

4.8 Nutrient use efficiency

Farmers are mostly concerned with the total profit and the marginal benefit-cost ratio from investment in fertilizers and soil conservation measures. The most relevant measure of nutrient use efficiency is, therefore, the partial factor productivity from investment in applied nutrients. The partial factor productivity from applied nutrients (PFP) is the ratio of grain yield to the quantity of applied nutrients.

Agronomic efficiency (AE)

It represents the marginal benefit from investment in nutrient inputs.

The formula for AE is:

$$AE = \frac{GY_{NA} - GY_{No}}{N_R}$$

where,

AE = Agronomic efficiency

GY_{NA} = Grain yield (kg/ha) with addition of nutrient

GY_{No} = Grain yield (kg/ha) without addition of nutrient

N_R = Rate of added nutrient (kg/ha)

Recovery efficiency (RE)

Crop management practices in general and nutrient management practices in particular have the greatest impact on the RE from applied nutrients. For example, competition from weeds, poor control of water, plant disease or insect damage can reduce the uptake of applied nutrients. Thus, improved crop management per se, including use of the best adapted variety, helps maximize the RE from applied nutrients.

Physiological efficiency (PE)

Physiological efficiency (PE) represents the ability of a plant to transform a given amount of acquired nutrient into economic yield. For cereals, economic yield is grain yield. There are significant differences in PEs among genotypes. Most of the variations are associated with differences in harvest index (HI) which is the ratio of grain yield to total plant biomass, for example, traditional rice varieties typically have an HI of 0.33 while modern high yielding semi-dwarf rice varieties (MV) have an HI of about 0.5. Therefore, with an equivalent quantity of N uptake and total biomass production MV will have greater PE than the traditional variety.

4.9 Trend of fertilizer use in Bangladesh

Inorganic fertilizers have been introduced in this country during early 1950's as a supplemental source of plant nutrients. But their use steadily increased from the mid 1960's along with the introduction and expansion of modern high yielding varieties of crops accompanied by the development of irrigation facilities. The increasing trend of fertilizer use, particularly urea-N, still continues (Appendix-1). Until 1980, three primary major plant nutrients (N, P & K) along with one secondary major nutrient (Ca) were supplied from fertilizer to this country's soils. The importance of S and Zn for rice culture in particular was recognized during early 1980's. Gypsum and zinc sulphate, respectively were then introduced to supply these nutrients. Very recently, the deficiencies of Mg, B and Mo have been reported for some soils and crops. Of the total nutrients

used in the country, N alone constitutes about 80 percent, which may lead to nutrient imbalance in soil-plant systems. If this trend of fertilizer use continues along with intensive cultivation of high yielding crop varieties, the productivity of soils is likely to be seriously affected. Factor productivity of nutrients has already been declined. To avert this potential danger, the limiting nutrients must be identified and the soils should be enriched with the addition of those nutrients in a properly balanced fertilization programme. Prior to 1990, only TSP was used by the farmers as a source of P because it was the only P containing fertilizer available in the market that time. After 1990, DAP, another source of P, had been made available in the market. Since these materials vary in their P content, variable amounts of these materials would be required to meet a specific P fertilizer recommendation. It is to be noted that SSP contains S and DAP contains N in addition to P. Therefore, there is a need to adjust for S and N fertilizer application if either SSP or DAP is used as a source of P instead of TSP. A list of commonly used fertilizers with their nutrient content is given in Appendix-2.

Looking at the fertilizer consumption over the years from 1980-81 to 2020-21, use of every nutrient (N, P & K) has increased as the time has advanced, except during 2006-2008 in which time cost of fertilizer became too high which the farmers could not afford (Fig. 4). The N consumption in 1980-81 was 365881 tons which in 2015-16 increased to 1183024 tons. Between P and K, the trend of K use is higher than that of P use. Fertilizer consumption has much increased since the government has given subsidy on fertilizer price (urea, TSP, DAP, MOP and Gypsum).

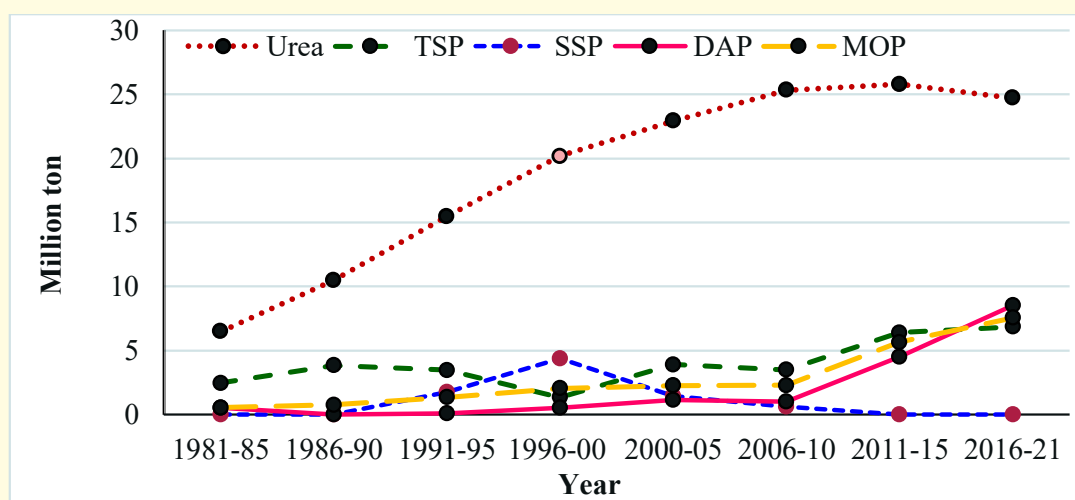


Fig. 4 Consumption of N, P and K fertilizers in Bangladesh during 1981 to 2021. (Annual N+P+K use: Bangladesh 320 kg/ha, India 170 kg/ha, world average 200 kg/ha)

4.10 Negative effects of excess use of fertilizers

Fertilizer application is a prerequisite to supply nutrients (when deficient in soil) to plants for achieving higher and sustained crop yield. Nevertheless, the use of chemical fertilizers can have negative impacts on soil, water and crops particularly when an excess amount is used. Excess use of fertilizers causes excess uptake of nutrients producing toxic effect on crops. Their residual and unused amounts will become pollutants for air, water, and soil. Overuse of fertilizers can cause soil acidification, soil crust and GHG emissions.

Long-term use of chemical fertilizers viz. urea can decrease soil pH, which in turn affects bacterial activity. Excess and/or continuous use of fertilizers may lead to heavy metal accumulation, eutrophication and accumulation of phosphate and nitrate in plants. Eutrophication is the process by which water body (pond, lake) becomes progressively enriched with mineral nutrients, particularly nitrogen and phosphorus which encourages the growth of algae (algal blooms). Triple superphosphate fertilizer may contain heavy metals such as cadmium and arsenic that may accumulate in soils, plants and enter human body via food chains. Thus, excessive use of fertilizers is detrimental to crop yield, food quality, soil microbial growth and pond water quality (eutrophication, heavy metals pollution). The adverse effects of these synthetic chemicals on human health and the environment can be reduced or eliminated by adopting new agricultural technological practices, preferably following GAP protocols including the use of organic inputs such as manure, biofertilizers, slow-release fertilizers, nanofertilizers, etc.

4.11 Identification of adulterated fertilizers

The most important thing in quality control is the identification of adulterated fertilizers. Simple methods of identification of adulterated fertilizers practicable at the field level have been developed by the SRDI. Through long experience, the SRDI has identified various materials used in adulteration of fertilizers. These materials can easily be identified through qualitative analysis of fertilizer samples using some locally available materials. Following these methods, a preliminary idea can be obtained about the quality of fertilizers. To know the degree of adulteration, quantitative analysis must be done in the well-equipped laboratories.

Identification of adulterated urea

Adulteration of urea is very rare. In few cases misbranding *i.e.* packaging of other fertilizer in the bag of urea is observed. Urea fertilizer never exists in crystalline form.

For quality testing, pour one teaspoonful of urea into two teaspoonful (10 ml) of water and stir them. Urea will be dissolved quickly and a clear solution will be developed. The solution will be felt cool as urea is hygroscopic. If other materials like lime are mixed with urea it will produce a pungent odour of ammonia gas when dissolved in water.

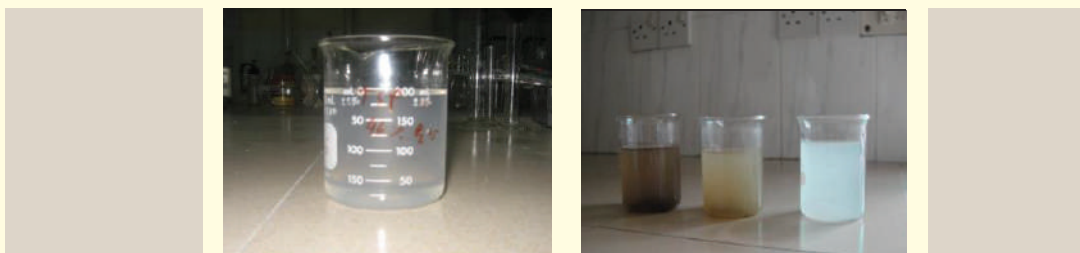


Good quality urea solution

Identification of adulterated triple super phosphate (TSP)

Triple Super Phosphate is a widely used phosphatic fertilizer in Bangladesh. It is granular, grey to dark grey in colour and contains 20% total P and 17.4% water soluble P with acidic taste. A good quality TSP is strong acidic in taste with pungent smell but adulterated TSP has no acidic taste and pungent smell.

Good quality TSP is little bit hard and cannot be broken by pressing with two thumb nails. On the other hand, adulterated TSP can be broken easily by pressing between the thumb nails. Colour of good quality TSP is found homogenous when broken into pieces, while the colour of adulterated TSP is not homogenous when broken.



Good quality (Left) and adulterated TSP solution

To know about the quality, pour one teaspoonful of TSP in 100 ml of water in a beaker or glass and stir them for a few minutes. All the granules of TSP will be dissolved and a clear solution like green coconut water will be developed. On the other hand, a hazy/turbid/muddy solution is developed when adulterated TSP is dissolved in water.

Identification of adulterated diammonium phosphate (DAP)

DAP is a widely used chemical fertilizer in the country, which contains 18% N and 20% total P and 17.8% water soluble P. The DAP is highly soluble in water, having pungent smell and acidic in taste. It is granular and dark grey to white in color. The DAP becomes wet when exposed to air.

Place one to two teaspoonful of DAP fertilizer on a dry paper and keep it open for one to two hours. If it becomes wet, the fertilizer is of good quality but if it does not wet the DAP is adulterated. For quality testing, pour one teaspoonful of DAP into 100 ml of water in a beaker or glass. The good quality DAP will be dissolved quickly. After dissolution, add half teaspoonful of barium chloride (BaCl_2) into the solution. In case of good quality DAP, BaCl_2 will be settled down at the bottom of the beaker or glass. On the other hand, adulterated DAP will produce turbid/muddy/hazy precipitate in the solution due to presence of sulphur.



Good quality (Left) and adulterated (Right) DAP solution

Identification of adulterated muriate of potash (MOP)

The MOP is one of the most widely used potash fertilizers in Bangladesh. It is light to deep red in colour, crystalline in nature and contains 50% K. It does not have pungent smell or taste.

To know about the quality, pour half teaspoonful of MOP fertilizer into 100 ml of water in a beaker or glass and stir for a few minutes. The good quality MOP will be dissolved totally and almost a clear solution will be produced. But if MOP contains sand, powder of broken glasses, finely ground white rocks/stones or brick chips, these materials will not dissolve and will be settled at the bottom of the container or beaker. This indicates that the fertilizer is adulterated.

If MOP is mixed with synthetic dye (red or any other color), there will be a layer of colour at the upper surface of the solution when dissolved in water. In such cases, undoubtedly, the MOP is adulterated. The colour will stick on to the finger when dipped into the solution. The good quality



Good quality (Left) and adulterated (Right) MOP solution

MOP will produce light homogenous red colour solution. The colour will not stick on to the finger if deep into the solution.

Identification of adulterated gypsum

Gypsum is used as a source of S in Bangladesh. It comes out as by-product during production of TSP. Gypsum is amorphous/powder, grey to white in color, contains 16% S and 20% Ca. Gypsum contains comparatively high moisture and for that reason it cannot be preserved for a long time in open place. Gypsum fertilizer is soft and glitters when exposed to sunlight. Usually in rare cases adulteration is observed in gypsum, because of its lower price compared to other fertilizers.

To check its quality, pour one teaspoonful of gypsum into a small beaker or glass or ceramic pot and add 10-15 drops of dilute (10%) HCl. If there is effervescence of CO₂, then undoubtedly the gypsum is adulterated. Good quality gypsum will not produce gas or effervescence. Good quality gypsum is heavier than the adulterated one.



Good quality Gypsum

Identification of adulterated zinc Sulphate, heptahydrate (ZnSO₄.7H₂O)

Zinc sulphate, heptahydrate is crystalline like sugar. It contains 21% Zn and 10.5% S. The Government of Bangladesh has also approved granular zinc sulphate, heptahydrate for marketing and use in Bangladesh. Good quality zinc sulphate, heptahydrate must be in crystalline form and friable. To know about the quality, pour one teaspoonful of zinc sulphate, heptahydrate in 100 ml water in a glass container or beaker and stir. The good quality zinc sulphate will be totally dissolved and no sediment will be found at the bottom of the beaker or glass container. But in case of adulterated zinc sulphate, heptahydrate there will be sediment at the bottom of the beaker or glass container. Zinc sulphate, heptahydrate is lighter than zinc sulphate, monohydrate in weight.



Good quality ZnSO₄
(Hepta-hydrate) solution

Identification of adulterated zinc Sulphate, monohydrate (ZnSO₄.H₂O)

Zinc Sulphate, monohydrate must be granular and contains 36% Zn and 17.5% S. Good quality Zinc Sulphate, monohydrate is pale white in color but adulterated zinc sulphate may be milk white in colour. To know the quality of zinc sulphate, monohydrate, pour one to two teaspoonful of the fertilizer into 100 ml cool water in a glass container or beaker and stir. The fertilizer will not be dissolved fully and will produce turbid solution.

For quality testing, dissolve one to two teaspoonful of zinc sulphate, monohydrate into 100 ml of water in a beaker or glass container, at first some cotton like materials (sediment) will be seen in the whole solution. Allow it for 5-10 minutes, layer of sediment/ precipitate will be seen at the surface of the solution. Now add small amount (2g) of sodium carbonate or sodium bicarbonate. A dark turbid solution will be produced. After 5-10 minutes the dark turbid solution will become clear starting from the bottom to the top in case of good quality zinc sulphate, monohydrate and a white layer will be seen



Good quality (Left) and
adulterated (Right) ZnSO₄
monohydrate solution

at the surface. But in case of adulterated fertilizer, precipitation of MgCO_3 will be seen at the bottom of the beaker or glass container.

Identification of adulterated chelated zinc

Chelated zinc is a high quality zinc fertilizer. It is a compound of zinc and organic salt containing 10% Zn. Good quality chelated zinc is yellowish powder or fine crystalline in form and very light in weight.

For quality testing, pour one teaspoonful of chelated zinc into 200 ml of cool water in a beaker or glass and stir it. The fertilizer will be dissolved quickly and will produce a clear solution. Add little amount of barium chloride to the clear solution. There will be no precipitation or sedimentation since there is no sulphur. In case of adulterated chelated zinc, when it is poured into cool water, it will not be dissolved quickly. Upon addition of barium chloride white precipitate or sediment will be produced.



Good quality (Left) and adulterated (Right) chelated Zinc solution

Identification of adulterated boron fertilizers

Boric acid and Solubor are used as boron fertilizers in Bangladesh.

Boric acid is white, fine crystalline, contains 17% B, dissolves fully in cold water and leaves no sediment in solution at the bottom.

Solubor is white, fine powder, light weight, contains 20% B, dissolves fully in cold water and leaves no sediment in solution at the bottom.

For quality testing of 'Boric acid' and 'Solubor', pour one teaspoonful of boric acid or solubor into a beaker or glass containing 100 ml clear cold water and stir. Both the fertilizers will be dissolved quickly and will produce clear solution with no precipitation. Add small amount (2 g) of barium chloride into the solution. If the fertilizers are adulterated with 'sodium sulphate' (usually used), a white turbid/sediment like milk will be produced.

Boric acid is heavier in weight than solubor.



Adulterated (Left) and good quality (Right) Boric acid solution



Adulterated (Left) and good quality (Right) Solubor solution

Identification of adulterated organic fertilizer

Organic fertilizers are produced through decomposition of organic materials of plant and animal origin. The government of Bangladesh has approved the standard specification of organic fertilizer with certain physical and chemical properties through a Gazette Notification by the Ministry of Agriculture on 02 April 2008. As per notification the organic fertilizer should be-

- non-granular in form
- dark grey to black in colour
- odorless
- devoid of bad smell

A good quality organic fertilizer does not form clod when press in hand. These are the qualitative/physical properties of a standard organic fertilizer. By checking these properties preliminary idea about the quality of an organic fertilizer can be obtained. But it is not possible to be sure about the quality without chemical analysis in the laboratory.

5. Climate Smart Soil Management Practices

5.1 Soil organic matter management

Soil organic matter comes from plant and animal remains. It influences the physical, chemical and biological properties of soil. It improves soil physical conditions viz. soil structure, water holding capacity, aeration, soil erosion etc. It's a storehouse of plant nutrients, chiefly N, P & S. It serves as a food and energy for beneficial organisms viz. N₂ fixing bacteria (e.g. *Rhizobium*, *Azotobacter*), earthworms. It is the 'life force of a soil'.

Organic matter status

A good soil should have at least 2.5% organic matter, but in Bangladesh most of the soils have less than 1.5%, and some soils have even less than 1% organic matter. As the time advances, organic matter content in soil declines. This is particularly true under high land and medium high land conditions. The long-term fertilizer trials indicate that in the rice-rice (anaerobic-anaerobic) cropping system, the soil organic matter has slightly increased (BRRI and BAU reports) and in the wheat-rice (an aerobic-aerobic) system the soil organic matter has rather decreased (BARI report). Hence, depletion of soil organic matter cannot be generalized across the country. Organic matter status of Bangladesh soils is categorized as very low (0.408 mha, 4.7% arable land), low (2.58 mha, 30.1%) and medium (5.08 mha, 59.2%) (Hasan et al., 2020).

Organic matter mineralization

Soil organic matter undergoes mineralization and releases substantial quantities of N, P, S and smaller amount of micronutrients. Application of organic residues returns mineral nutrients to the soil. The conversion of organic N, P and S to available forms occurs through the activity of microorganisms and is influenced by temperature, moisture, pH etc. The rate and extent of mineralization determine crop availability of nutrients from added organic materials. All the bio-forms use soil as their home or they live on organic matter and decompose it to simple products. These products are responsible for sustaining soil productivity and performing environmental regulatory function.